SOP: Azure File Share and Azure File Sync

Objective

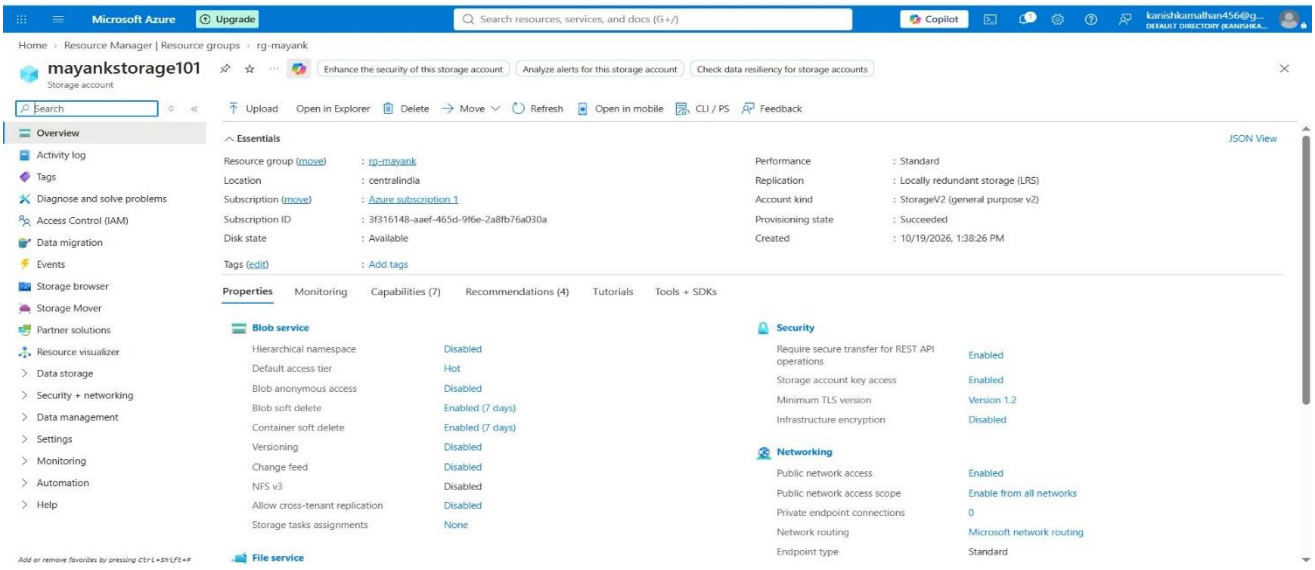
The objective of this lab is to create an Azure Storage Account and File Share, connect the file share to a Windows Server virtual machine, configure Azure File Sync, and verify file synchronization between the local server and Azure.

Step 1: Create a Storage Account

Open the Microsoft Azure Portal and navigate to **Storage accounts**. Click **Create** and configure the storage account with the name `mayankstorage101`, resource group `rg-mayank`, and region `Central India`. Select Standard performance and Locally-redundant storage (LRS) replication. Review the configuration and click **Create** to deploy the storage account.

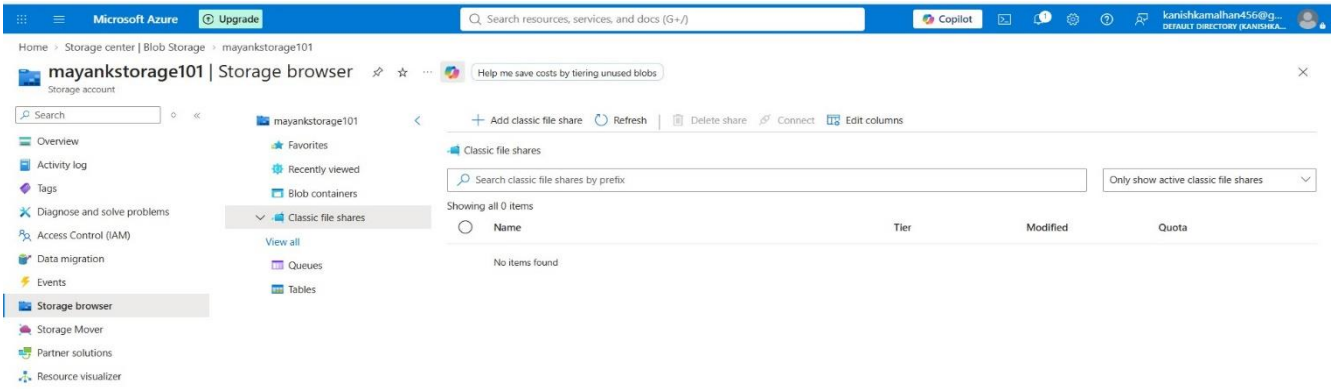
Step 2: Verify the Storage Account

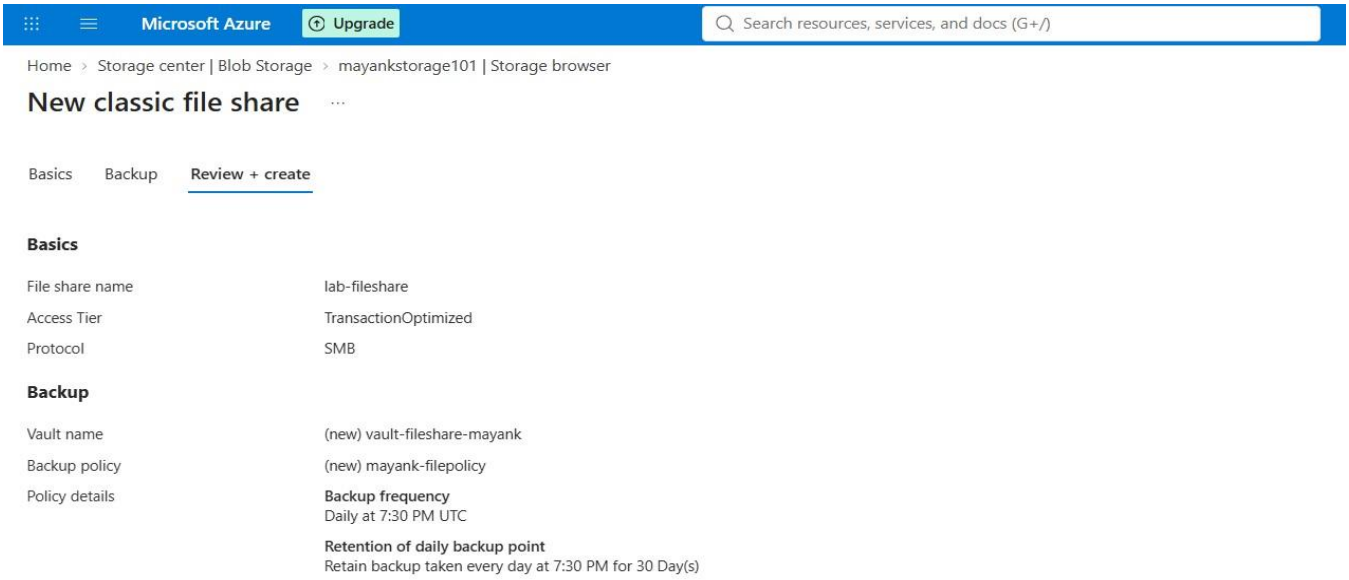
Open the newly created storage account `mayankstorage101`. Review the Overview page and verify the resource group, subscription, location, and storage account configuration.



Step 3: Create an Azure File Share

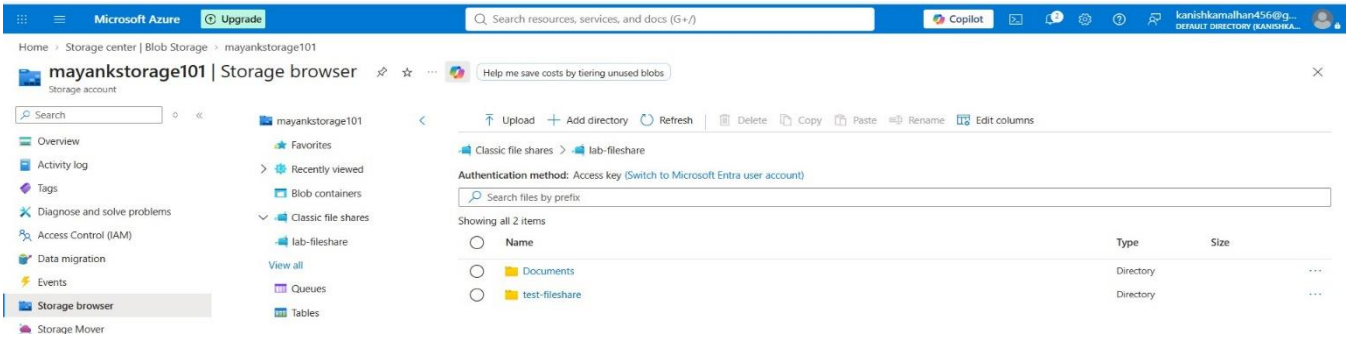
Open the storage account and navigate to **Data storage → File shares**. Click **+ File share**, enter `lab-fileshare` as the file share name, configure the required settings, and click **Create**.





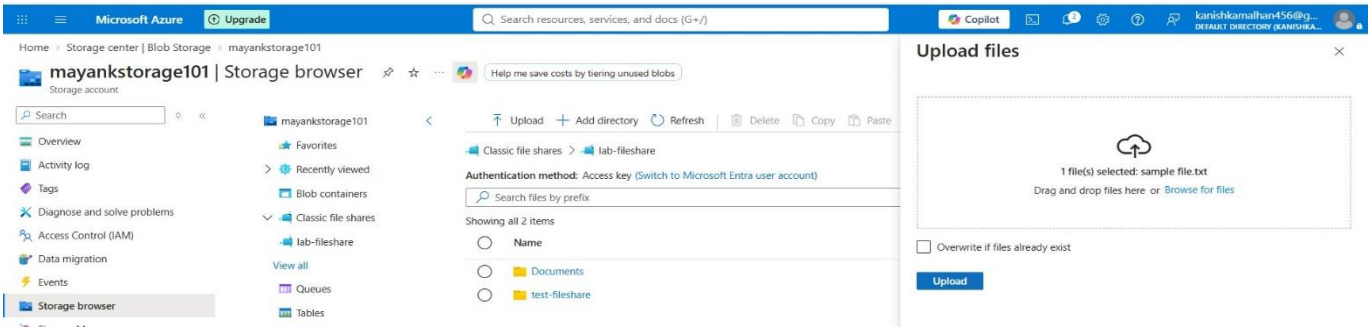
Step 4: Create Directories in the File Share

Open `lab-fileshare` and use the **Add directory** option to create the required folders, including `Documents` and `test-fileshare`. Verify that the directories appear in the file share.



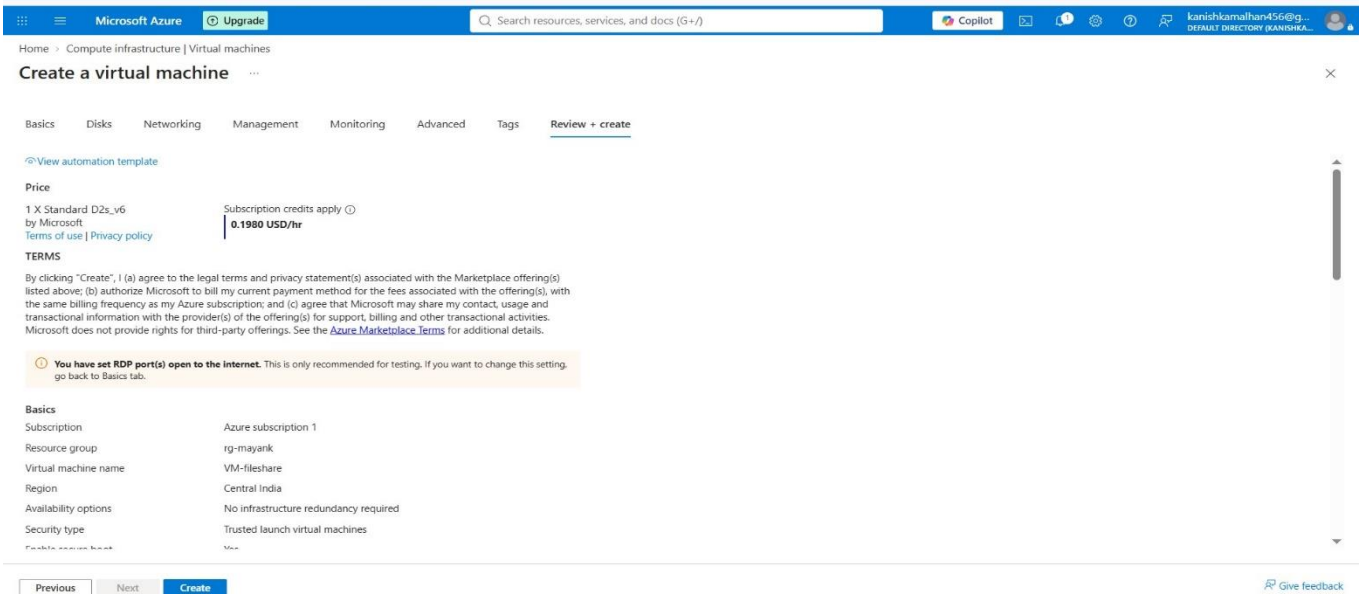
Step 5: Upload a File to the File Share

Open the required directory inside `lab-fileshare` and select **Upload**. Choose a sample file from the local computer and upload it. Verify that the file is visible in the Azure File Share.



Step 6: Create a Windows Server Virtual Machine

Navigate to **Virtual machines** in the Azure Portal and select **Create → Azure virtual machine**. Configure the virtual machine with the name `VM-fileshare`, select Windows Server 2022 Datacenter, choose the required region and VM size, configure administrator credentials and networking, and deploy the virtual machine.

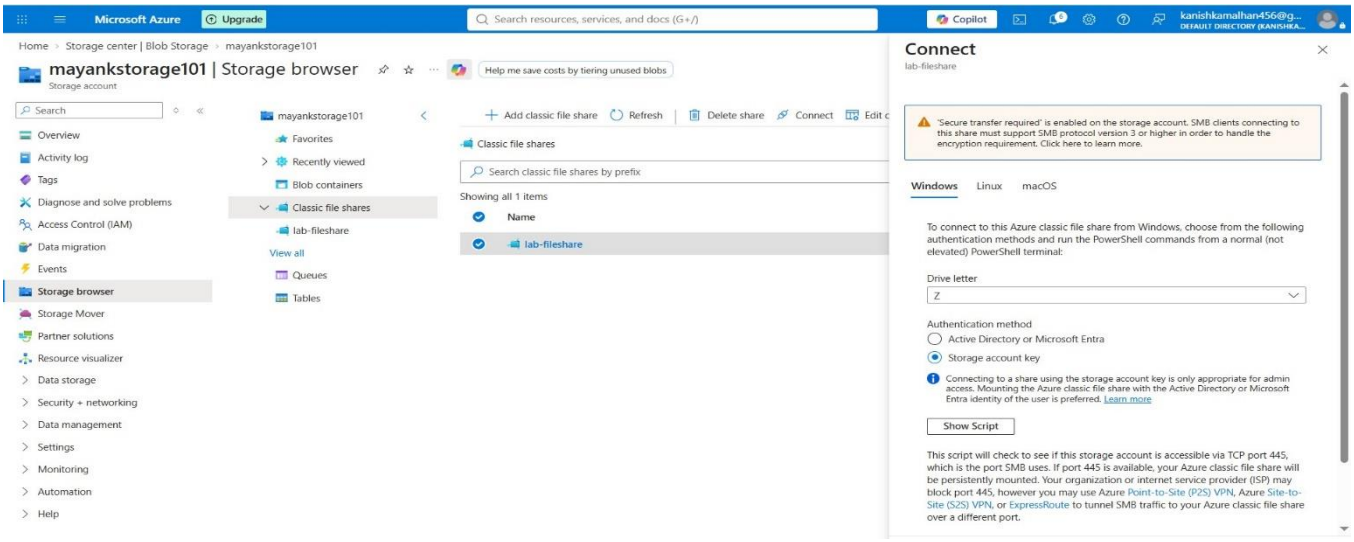


Step 7: Connect to the Windows Server VM

Open `VM-fileshare` in the Azure Portal and connect using Remote Desktop Protocol (RDP). Sign in with the administrator credentials and verify that the Windows Server desktop is accessible.

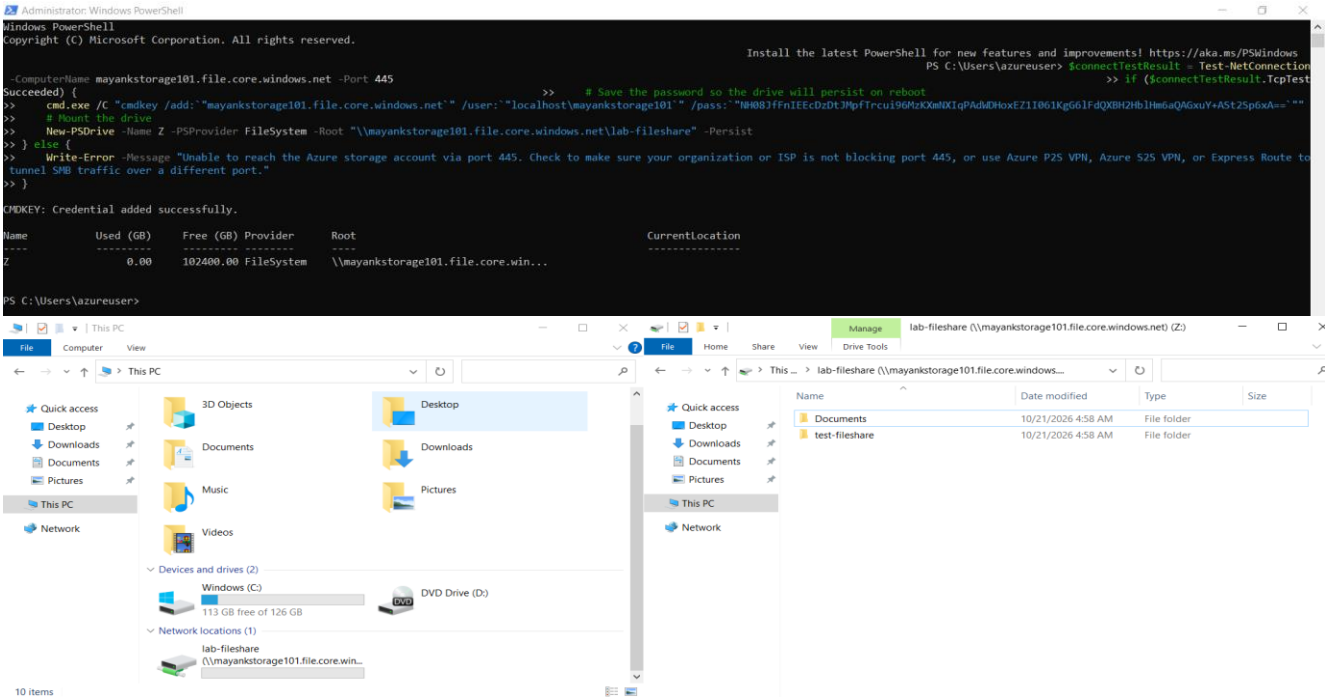
Step 8: Connect the Azure File Share to Windows Server

Open `lab-fileshare` in the Azure Portal and select **Connect**. Choose the Windows or PowerShell connection option and use the provided script to map the file share to the Windows Server as drive `Z:`. Run the script and verify that the connection is successful.



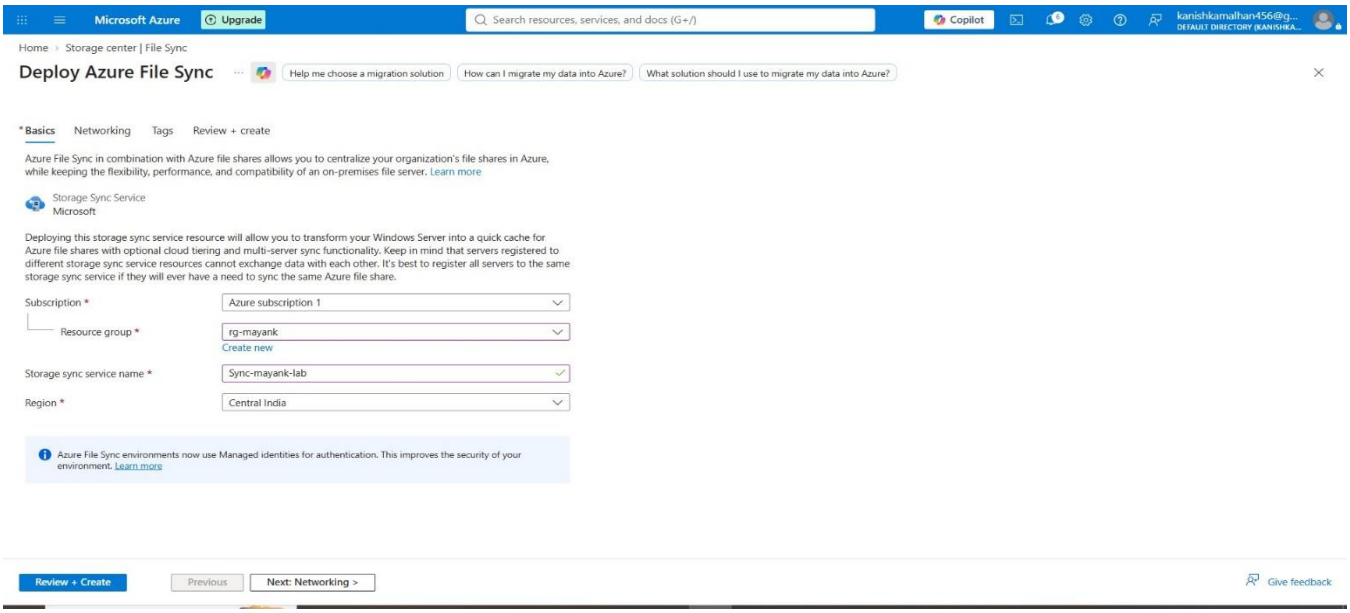
Step 9: Verify the Mapped File Share

Open File Explorer inside `VM-fileshare` and locate the mapped network drive `Z:`. Open the drive and verify that the `Documents` and `test-fileshare` folders are accessible.



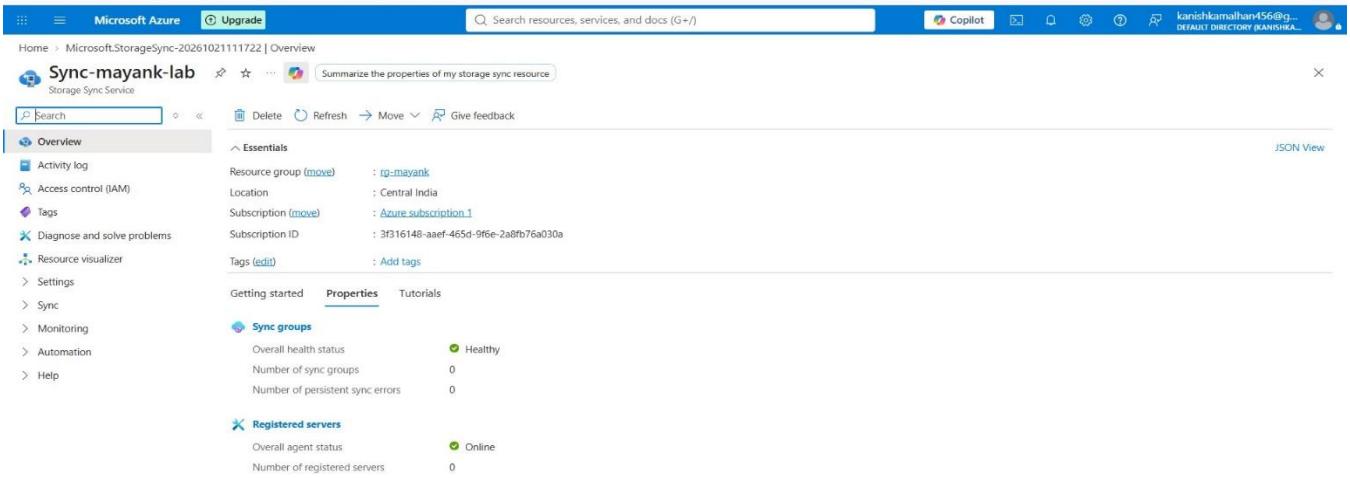
Step 10: Create the Storage Sync Service

Open the Azure Portal and navigate to **Storage Sync Services**. Click **Create**, select subscription `Azure subscription 1` and resource group `rg-mayank`, enter `Sync-mayank-lab` as the service name, and select the region `Central India`. Review the configuration and deploy the service.



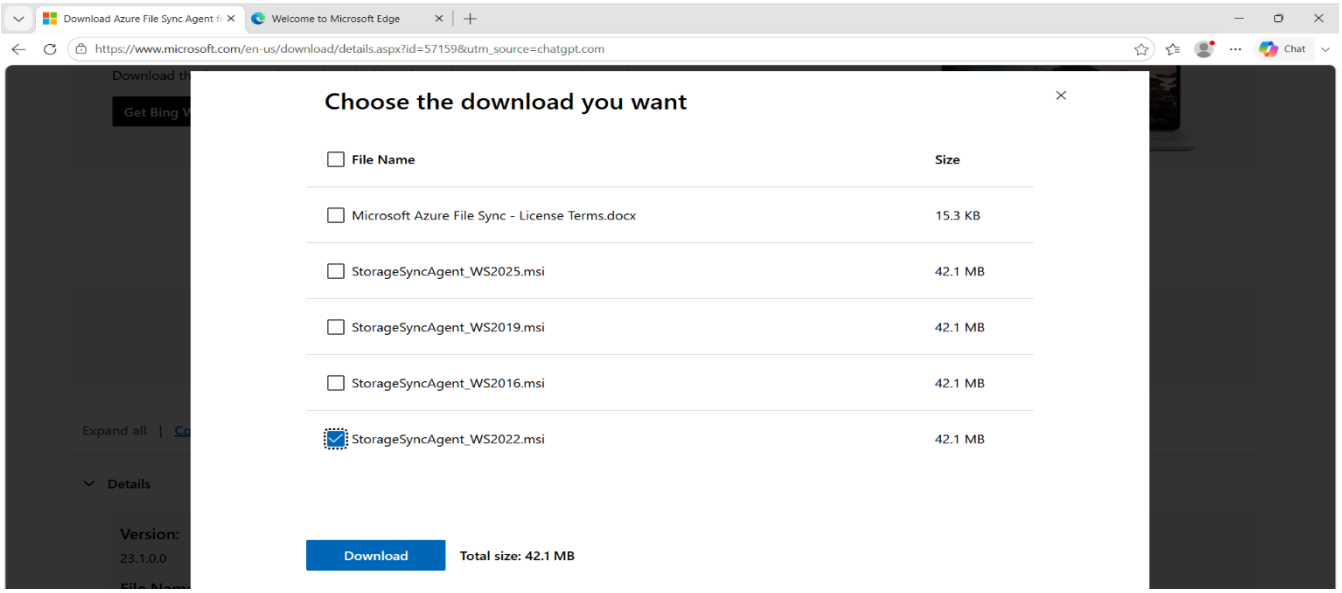
Step 11: Verify the Storage Sync Service

Open `Sync-mayank-lab` and review its Overview page. Verify the subscription, resource group, location, and health status to ensure that the Storage Sync Service has been created successfully.



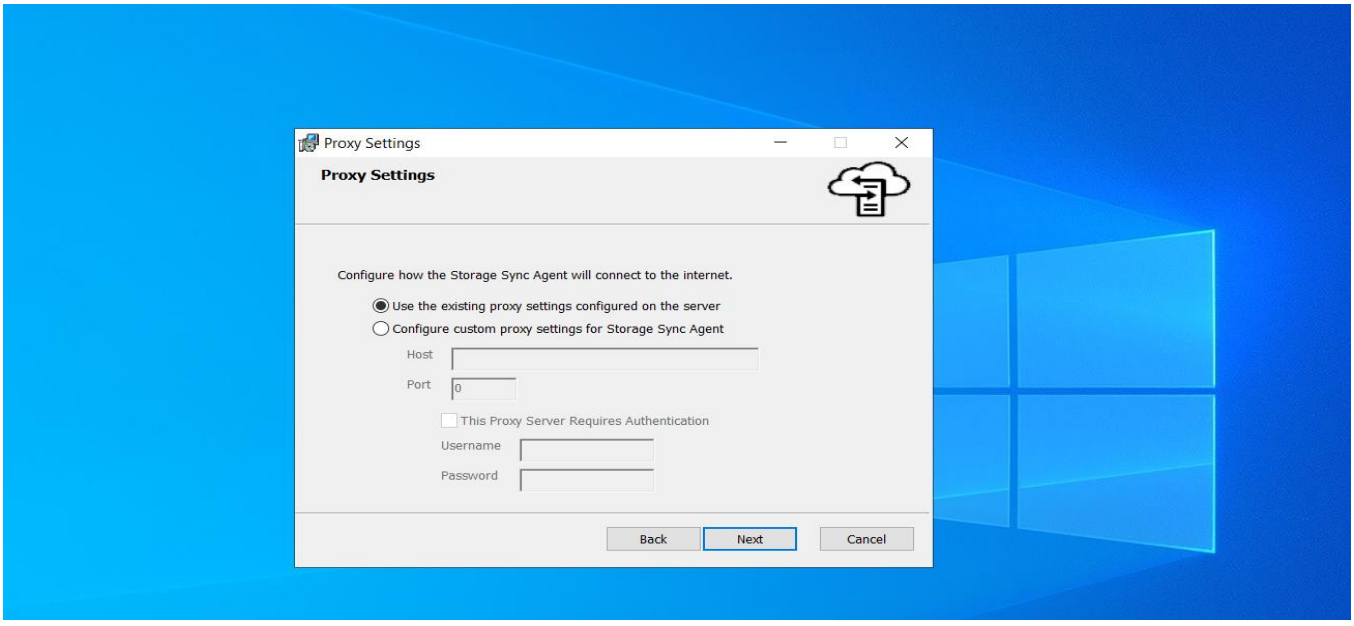
Step 12: Download the Azure File Sync Agent

Open the official Microsoft Download Center and locate the Azure File Sync Agent. Select `StorageSyncAgent_WS2022.msi` for Windows Server 2022 and download the installation package to the VM.



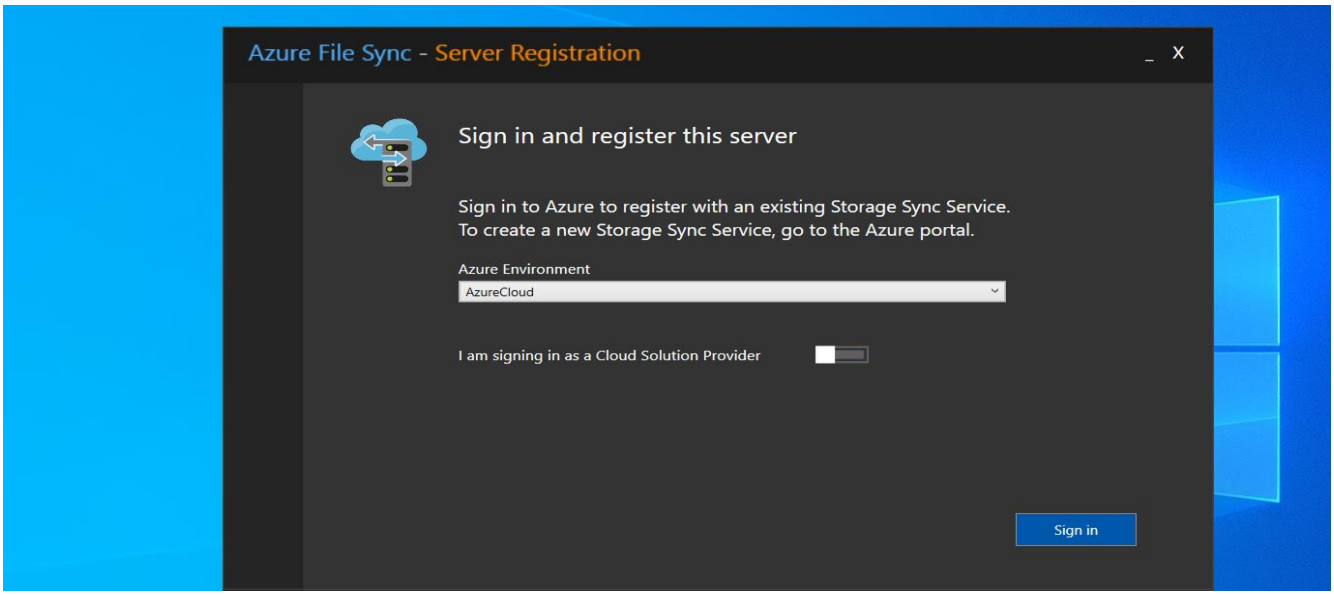
Step 13: Configure the Agent Installation

Run the downloaded MSI installer on the Windows Server. In the Proxy Settings window, select **Use the existing proxy settings configured on the server** and click **Next**. Continue through the wizard to complete the agent installation.



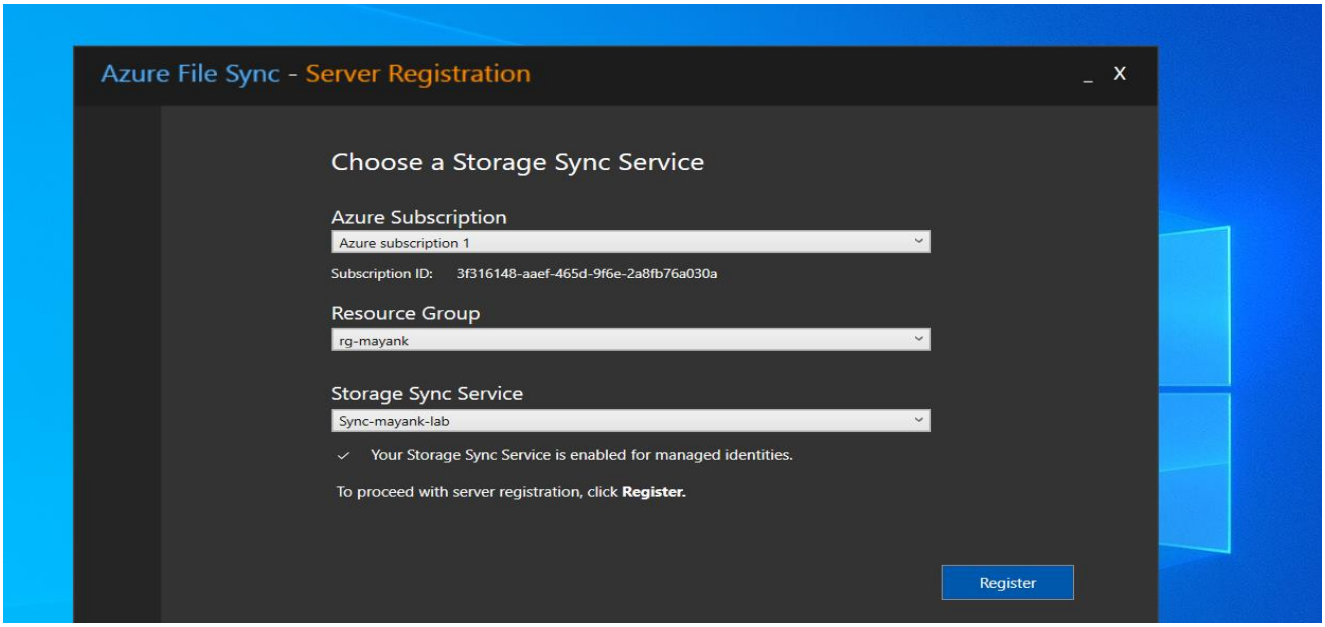
Step 14: Open Server Registration

After the agent installation, open the Azure File Sync Server Registration wizard. Keep the Azure Environment set to `AzureCloud` and leave the Cloud Solution Provider option disabled unless required by the environment. Click **Sign in** to authenticate with Azure.



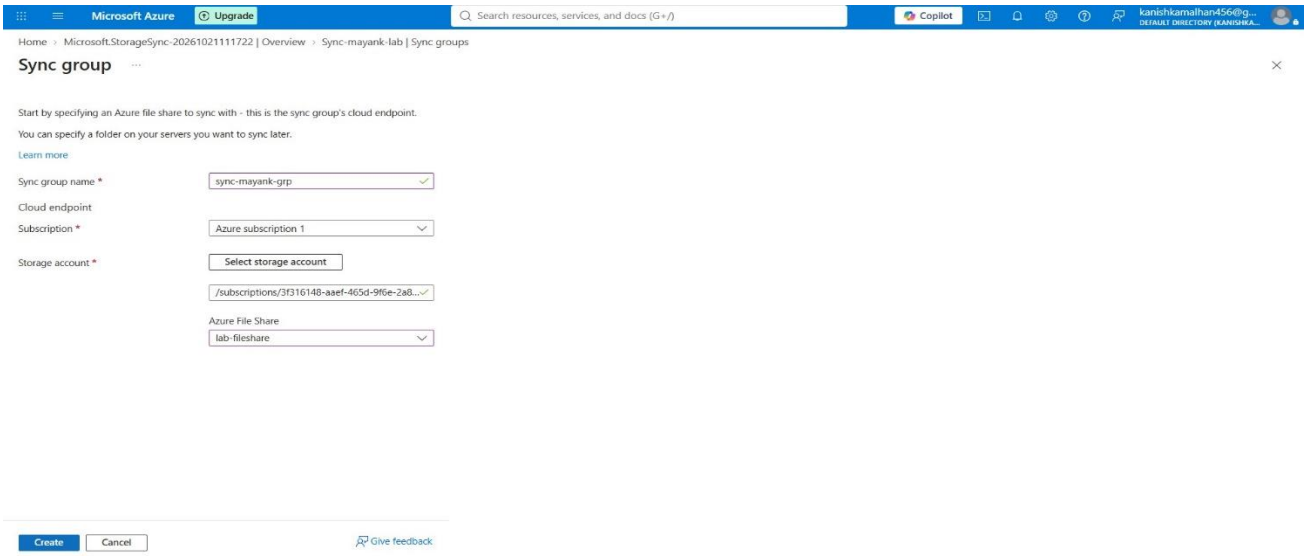
Step 15: Register the Windows Server

Sign in using an Azure account that has access to the Storage Sync Service. Select subscription `Azure subscription 1`, resource group `rg-mayank`, and Storage Sync Service `Sync-mayank-lab`. Click **Register** to register `VM-fileshare` with Azure File Sync.



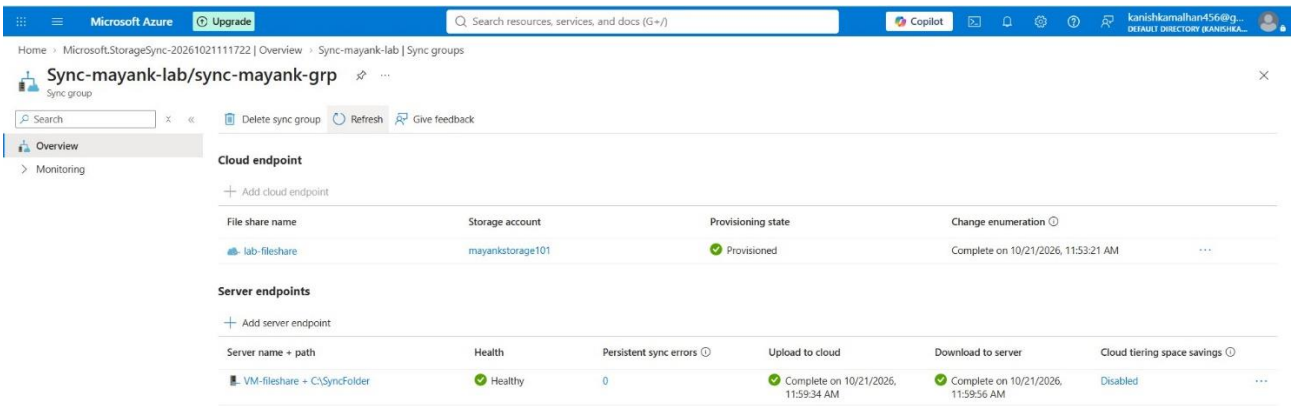
Step 16: Create a Sync Group and Cloud Endpoint

Open `Sync-mayank-lab` in the Azure Portal and navigate to **Sync groups**. Create a sync group named `sync-mayank-grp` and configure the cloud endpoint to use storage account `mayankstorage101` and file share `lab-fileshare`. Complete the creation process and verify that the cloud endpoint is provisioned.



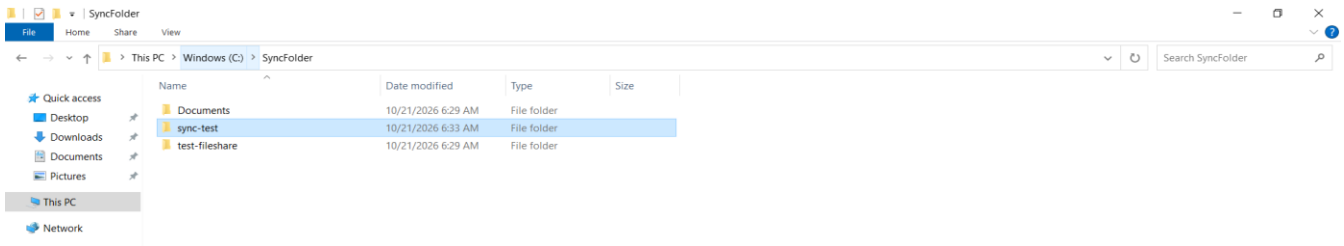
Step 17: Verify Cloud Endpoint and Server Endpoint

Open the created sync group and verify that the cloud endpoint lab-fileshare is in the Provisioned state. Under Server endpoints, verify that VM-fileshare is connected to the local path C:\SyncFolder. Check that the health status is Healthy, persistent sync errors are zero, and upload and download operations show Complete**.**



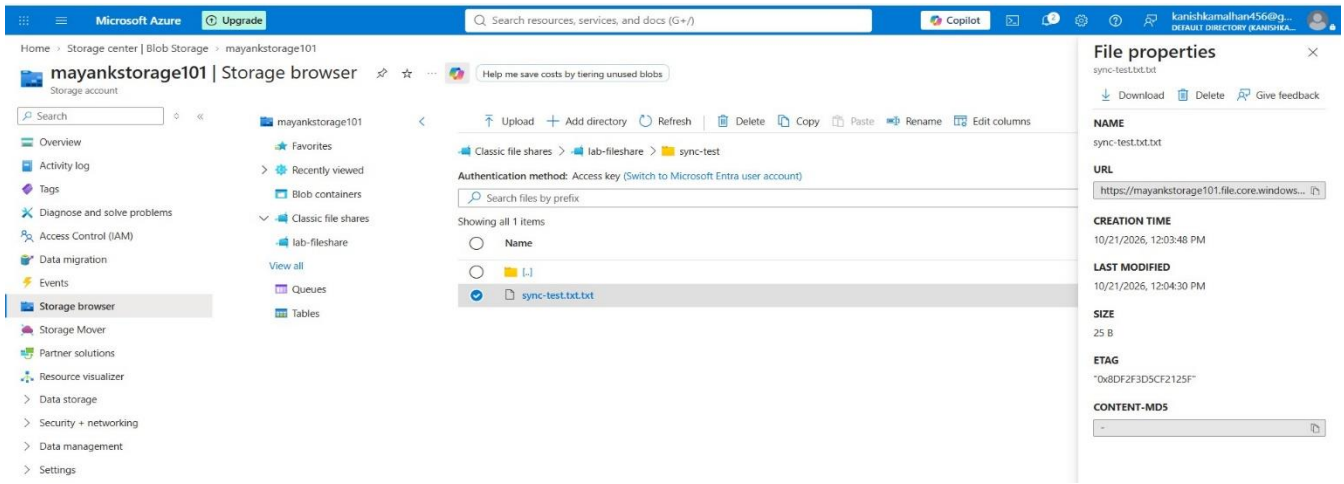
Step 18: Verify the Local Sync Folder

Connect to the Windows Server VM and open C:\SyncFolder in File Explorer. Verify that the folders Documents, sync-test, and test-fileshare are present. This confirms that the local folder is configured as the server endpoint for Azure File Sync.



Step 19: Verify Successful File Synchronization

After synchronization completes, check the Azure portal sync group status. Confirm that the server endpoint is healthy and that both Upload to cloud and Download to server show Complete. Verify the test file in the Azure File Share to confirm that data is available in the cloud endpoint.



Result

The Azure Storage Account `mayankstorage101` and File Share `lab-fileshare` were configured, and the file share was connected to the Windows Server VM `VM-fileshare`. Azure File Sync was configured using Storage Sync Service `Sync-mayank-lab`, sync group `sync-mayank-grp`, and server endpoint path `C:\SyncFolder`.